# CS 391
# Web Application Development

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**Week - 1**

# Outline

1. Development Environment
2. Web Architecture Overview
3. What are Web Standards?
4. HTML Introduction
5. HTML Forms
6. HTML Form Elements and Attributes

# Development Environment

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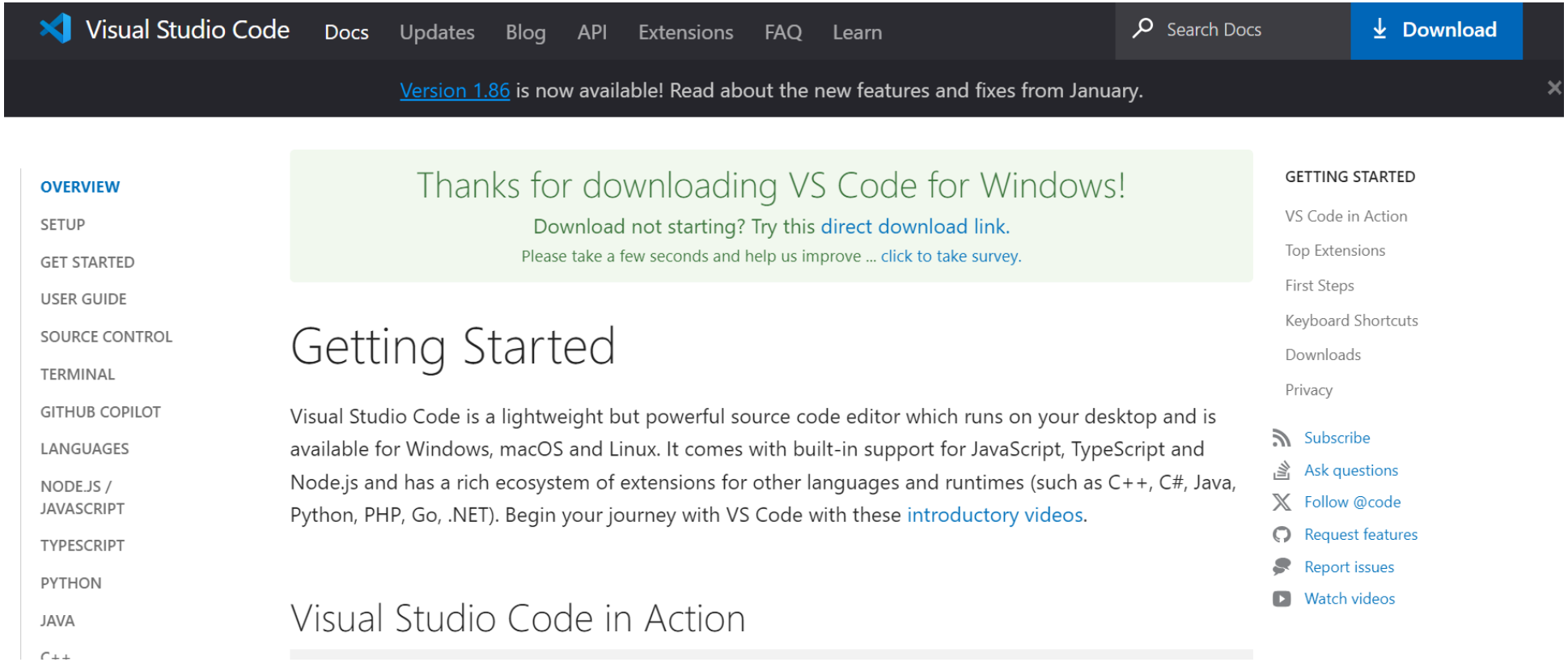
# Visual Studio Code

- https://code.visualstudio.com/download



# Postman

- https://www.postman.com/
- API Client
  - Quickly and easily send REST requests directly

# Git

- https://git-scm.com/
- Git is a free and open-source distributed version control system

# Web Architecture Overview

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# Web Architecture

- The Internet consists of **servers**, **clients**, and **routers**
  - **Clients**
    - A client requests files by sending a HTTP request to server
      - The request is sent over the internet using sockets
      - The file is specified in the request using a Uniform Resource Locator (URL)
    - Access information provided by the servers
    - Web browsers are the most common web clients
  - **Servers**
    - Web servers provide information to web clients
    - When a request comes in from a client, the server “serves” a response
    - The response contains header information as well as the content of the page
  - **Routers** provide the network that allows clients and servers to communicate

# HTTP Request

- The request is formed by the client to inform the server of the request

**http://www.myapp.com:8080/myapp/index.html**

URL: **U**niform **R**esource **L**ocator

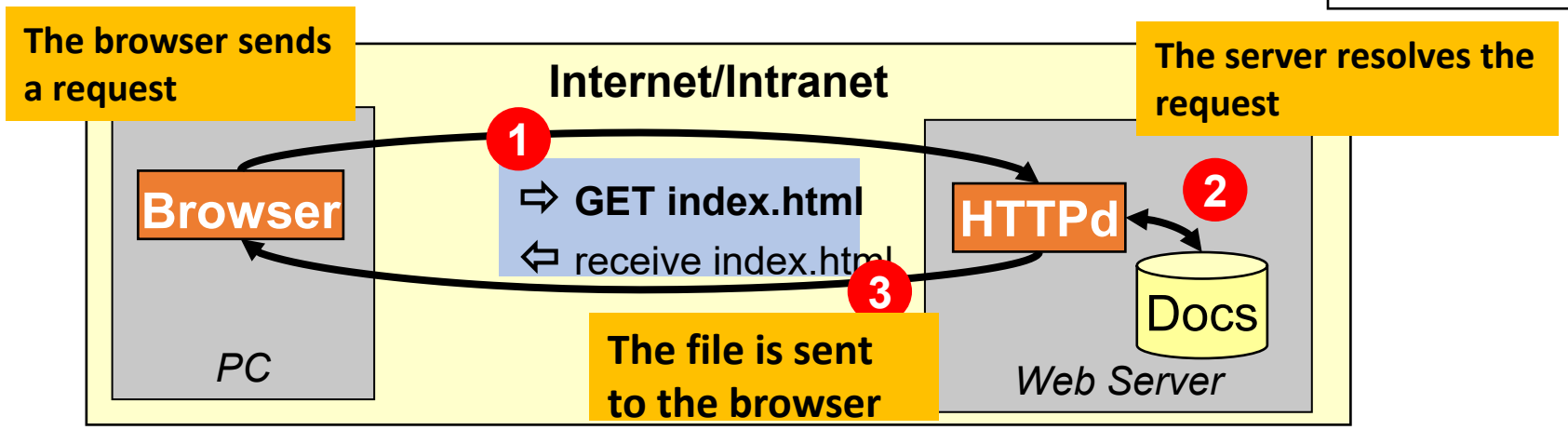
Name of server  Port  Application

name of page to download

```
Get /myapp/index.html HTTP/1.1
Accept: text/html
Accept-Language: en-us
User-Agent: Mozilla
```

**Request Header**

sent to “www.myapp.com:8080”



# HTTP Response

- The server's response includes a header followed by content data
  - The client uses information in the header to determine what to do with the content
- The response header includes
  - Content type, content length, cookies, ...

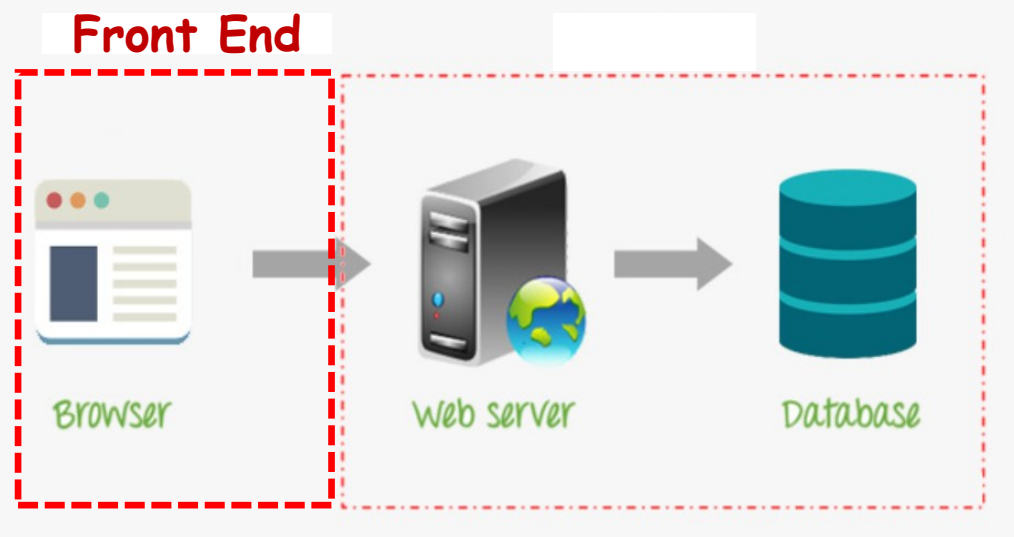
```
HTTP/1.1 200 OK
Date: Mon, 27 Jul 2020 12:28:53 GMT
Server: Apache/2.2.14 (Win32)
Last-Modified: Wed, 22 Jul 2020 19:15:56 GMT
Content-Length: 88
Content-Type: text/html

<html>
<body>
<h1>Hello, World!</h1>
</body>
</html>
```

index.html

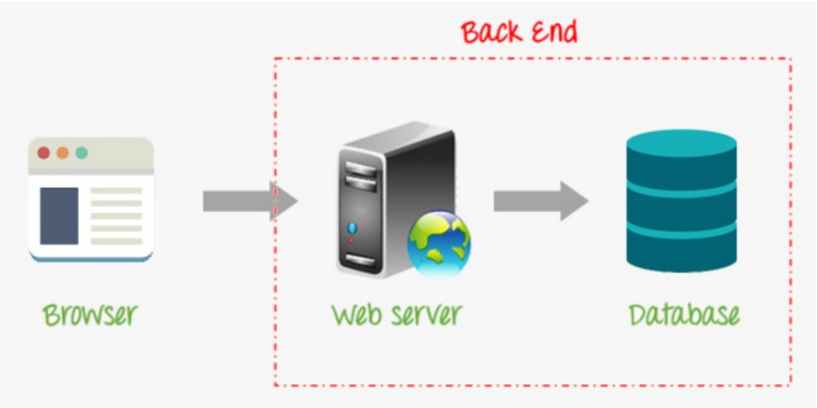
# Front-end Development

- Front-end development manages everything that users visually see first in their browser or application.
- Front-end developers are responsible for the look and feel of a site.
- A front-end developer works with the following technologies:
  - HTML, CSS, and JavaScript
  - Responsive design
  - Bootstrap
  - Frameworks/Libraries: ReactJs, VueJs, Angular, etc



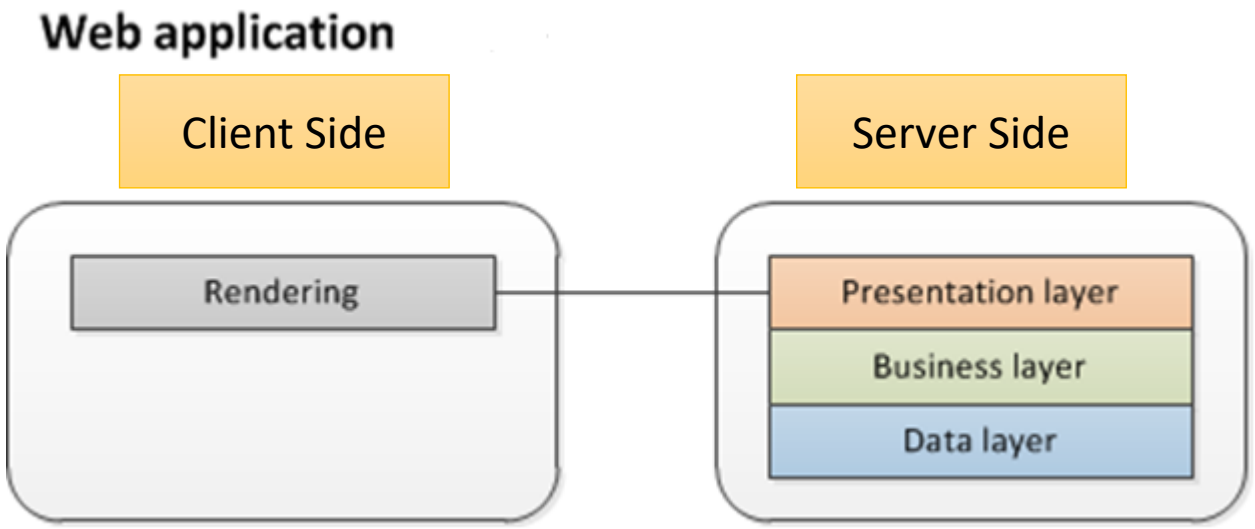
# Back-end Development

- Server-side development is used for the behind-the-scenes activities that happen when performing any action on a website.
- A backend developer works with the following technologies:
  - **Database :** Oracle, DB2, MySQL, MongoDB, SQLServer, etc
  - **Application Server :** Apache, WebLogic, WebSphere, etc
  - Web Services (REST & SOAP)
  - Server-side programming languages like Java, Python, Ruby, . Net, C++, JavaScript, PHP etc.



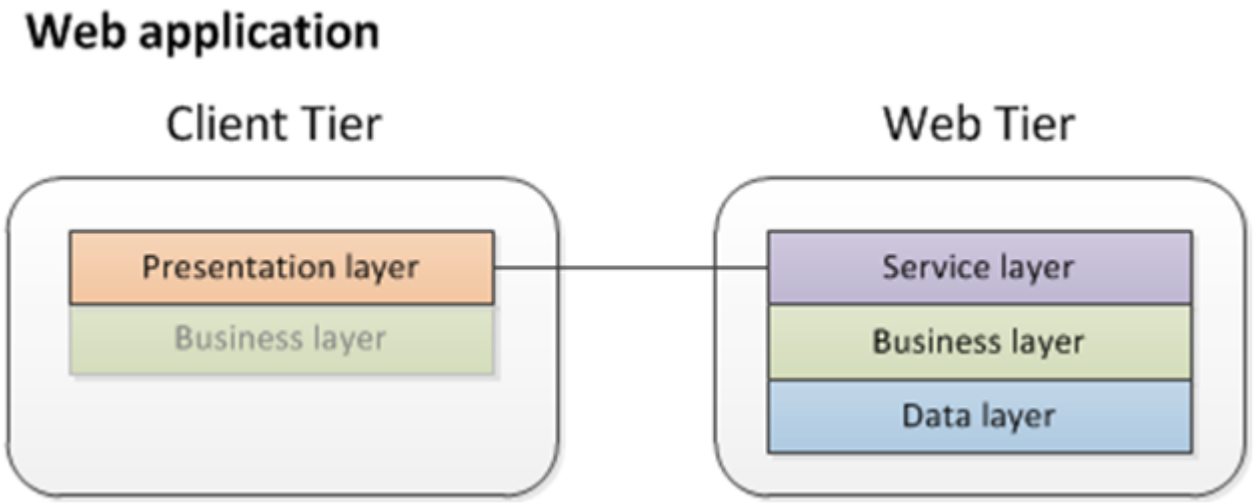
# Presentation Layer Architecture (old style)

- In an “old style” web application, the presentation layer is within the web tier, renders output to the client tier (browser)



# Presentation Layer Architecture (new style)

- Nowadays web sites are turning into Web apps
- Presentation layer moves to client tier
- May also have an element of business logic in client tier



# Presentation Layer Technologies

- Presentation layer technologies are HTML, JavaScript, CSS
- Easy to end up with spaghetti code – global variables, functions whose purpose is unclear, etc.
- Can improve this situation using patterns which have been tried and tested in the presentation layer in the Web Tier
  - Some client-side frameworks/libraries are:
    - Angular
    - React
    - Vue

# Web Standards

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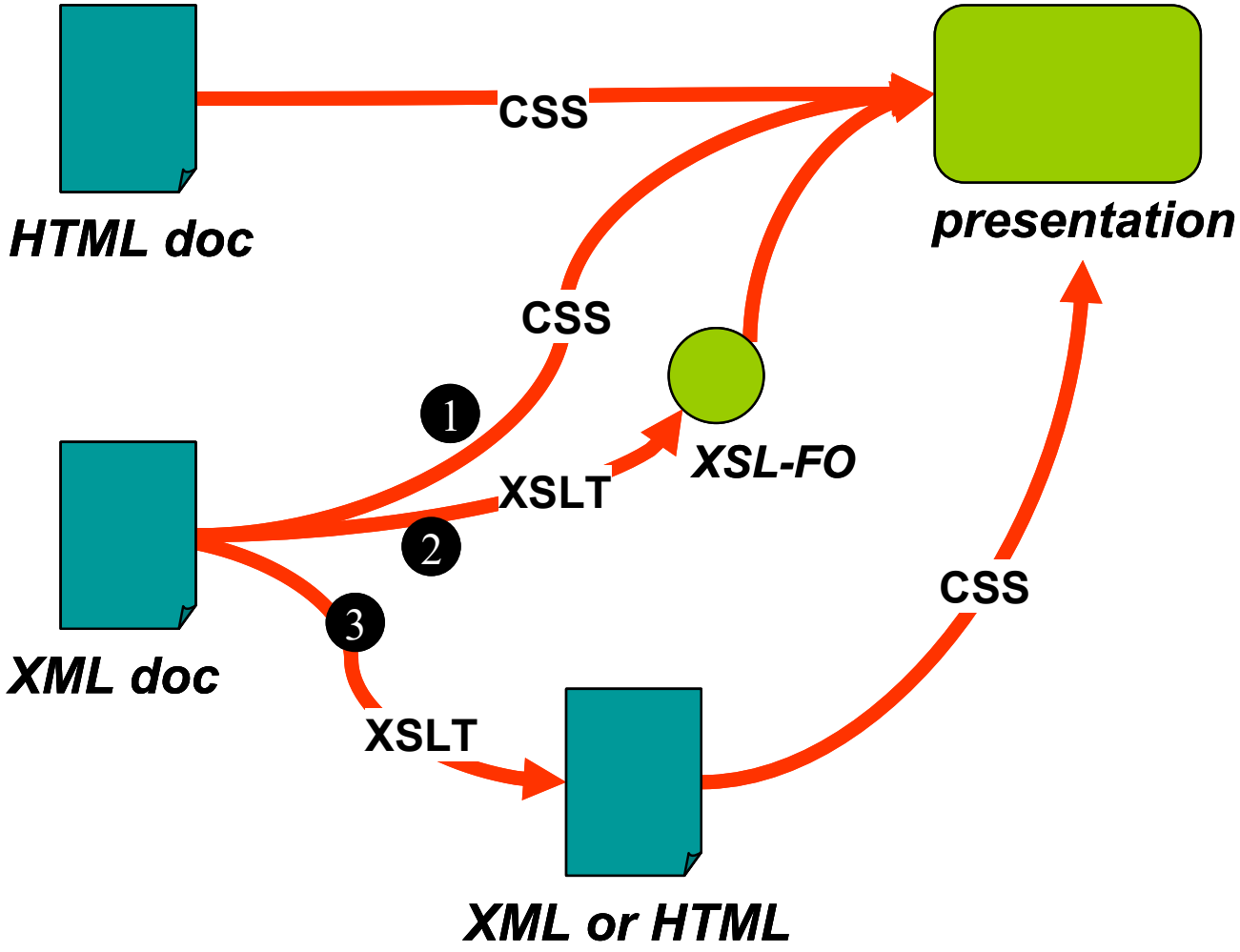
# What are Web Standards?

https://www.w3.org/

- Worldwide Web Consortium (W3C)
- Recommends Standards for Web Development
  - Structural Languages
    - HTML – Publishing Language of the Web
    - XHTML - Extensible Hypertext Markup Language
    - XML - Extensible Markup Language
  - Transformations
    - XSL - Extensible Stylesheet Language
    - XPath – XML Path Language
  - Presentation
    - CSS - Cascading Style Sheets

# Standards Related



# Web standards are important

- Designing and building with Web standards
  - Simplicity
    - Simplifies and lowers the cost of production
  - Accessibility
    - Delivers sites that are accessible to more people and more types of Internet devices.
  - Continuity
    - Sites will continue to function correctly  as traditional desktop browsers evolve, and as new Internet devices come to market

# HTML Introduction

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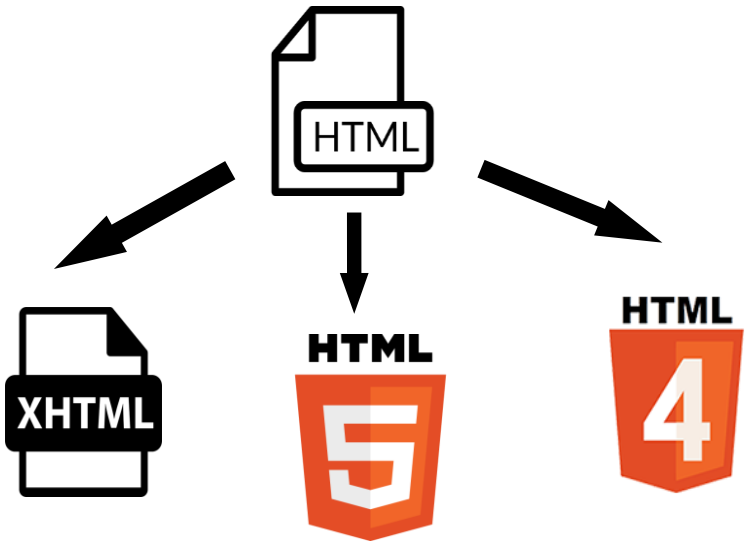
# HTML Overview

- **HTML: H**yper**T**ext **M**arkup **L**anguage
- HTML files consist of text and tags
  - Text provides the content of the page
  - Tags provide formatting "suggestions" to the client
- HTML tags are **case-insensitive**
  - Lowercase is an HTML5 Convention
- Whitespaces within HTML files are ignored
- HTML elements can be nested
  - start and end tags are properly nested.
    - Ex: text is not allowed.
    - text is allowed.

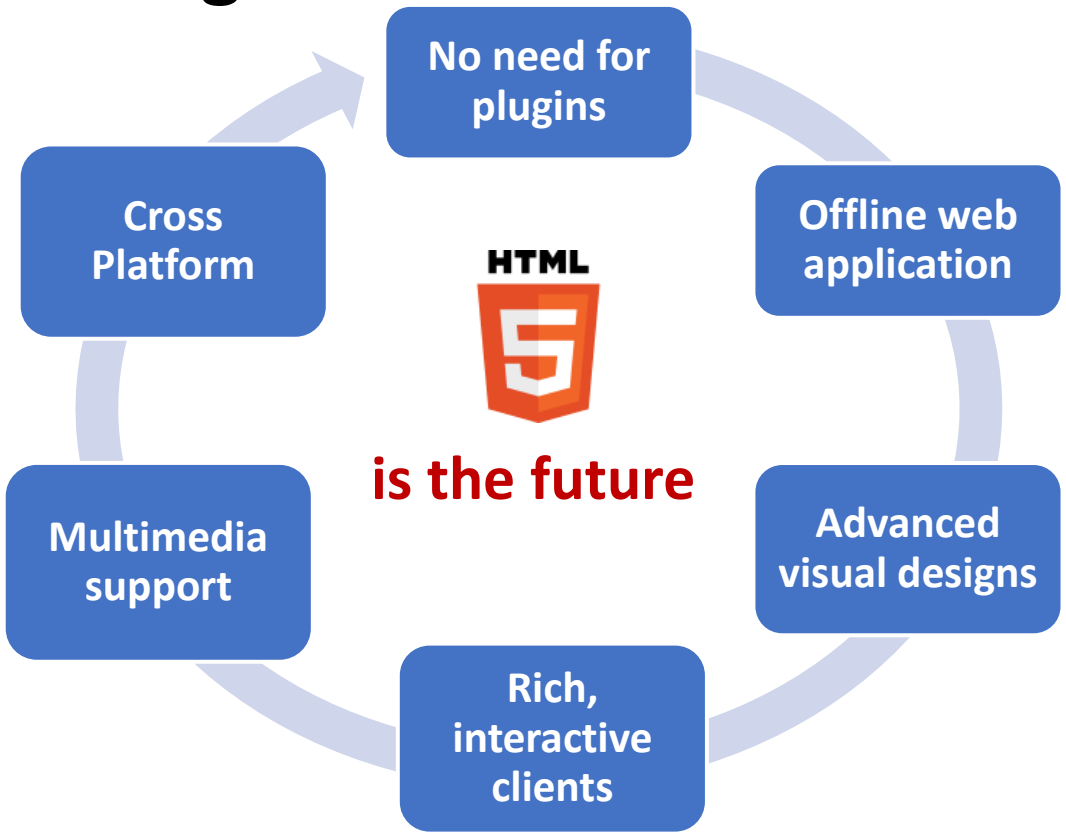


# What is HTML5?

- Latest version of Hypertext Markup Language
  - Currently under development
    - First draft spec at 2008
    - Far away from final version
  - Developed under the HTML 5 working group of the World Wide Web Consortium (W3C)

# HTML5 is not just "one thing"

- HTML5 is a **collection of technologies**
  - Structure by HTML
    - Revised and extended
  - Presentation by CSS
    - Revised and extended
  - Functionality
    - New APIs



# HTML5 Document Structure

<!doctype> declaration

Root element

```
<!DOCTYPE html>
<html>
<head>
      <meta charset="UTF-8">
      <title>Title of the document</title>
</head>
<body>
      …Content of the document…
</body>
</html>
```

# <!DOCTYPE> Declaration…

- All HTML documents must start with a <!DOCTYPE> declaration
  - Not an HTML tag
  - It is an "information" to the browser about what document type to expect.
- The <!DOCTYPE> declaration is NOT case sensitive

# …<!DOCTYPE> Declaration

- HTML 5:
  - <!DOCTYPE html>
- HTML 4.01
  - <!DOCTYPE HTML PUBLIC "-//W3C//DTD HTML 4.01 Transitional//EN" "http://www.w3.org/TR/html4/loose.dtd">
- XHTML 1.1
  - <!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.1//EN" "http://www.w3.org/TR/xhtml11/DTD/xhtml11.dtd">

# HTML Elements

- An HTML element is defined by a start tag, some content, and an end tag.
  - <tagname>Content goes here...</tagname>
  - <h1>My First Heading</h1>
  - My first paragraph

# Nested HTML Elements

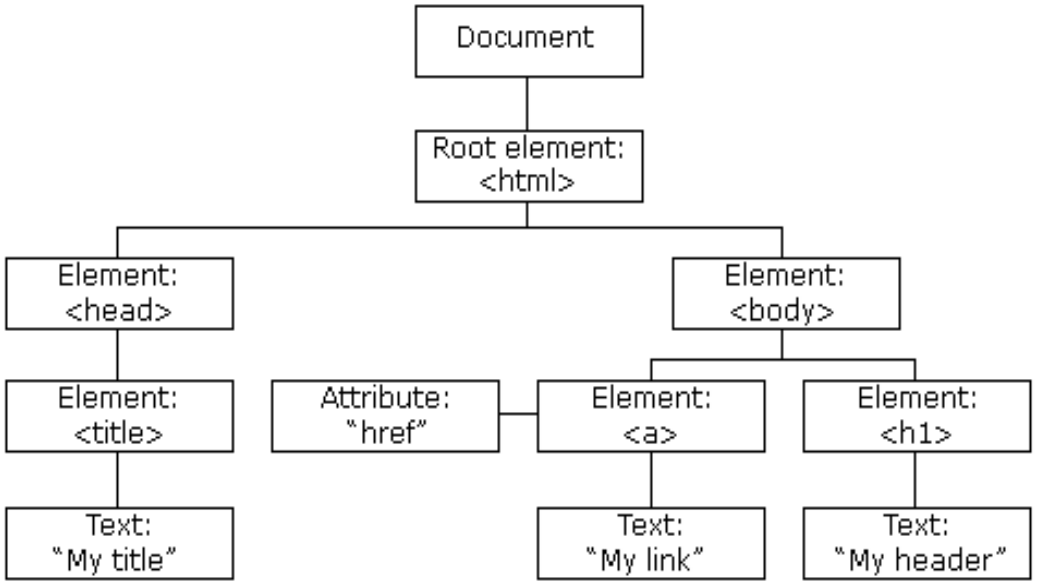
- Elements can contain other elements

```
<html>
  <head>
    <title>Hello html world</title>
    <meta charset="utf-8" />
  </head>
  <body>
    <h1>Introduction to html5</h1>
    <h3>Introduction to <b>tags</b></h3>
    <hr/>
    <a href="https://www.ozyegin.edu.tr/">Ozyegin University</a>
  </body>
</html>
```

# The HTML DOM

- **DOM:** Document Object Model
- Programming interface for HTML
- Defines a standard for :
  - The HTML elements as objects
  - The properties of all HTML elements
  - The methods to access all HTML elements
  - The events for all HTML elements
- When a web page is loaded, the browser creates a Document Object Model of the page.

# DOM Tree




```
<html>
  <head>
    <title>My Title</title>
  </head>

  <body>
    <h1>My header</h1>
    <a href="https://www.mylink.com.tr/">
       My link
    </a>
  </body>

</html>
```

# HTML Attributes

- All HTML elements can have attributes
- Attributes
  - provide additional information about elements
  - always specified in the start tag
  - usually come in name/value pairs like *name="value"*

Element

Start tag

End tag

```
<a href="https://www.ozyegin.edu.tr/" target="_blank">Visit Ozyegin University!</a>
```

attribute value attribute value

# HTML Tags - <meta>...</meta>

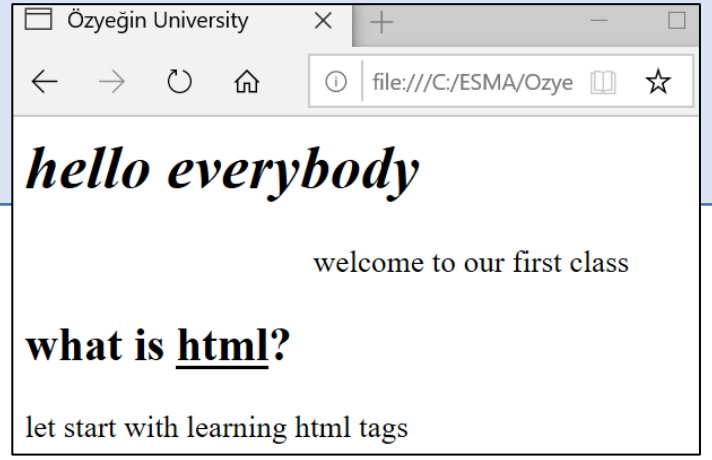
- Metadata is data (information) about data.

```
<head>
  <meta charset="UTF-8">
  <meta name="description" content="Free Web tutorials">
  <meta name="keywords" content="HTML, CSS, JavaScript">
  <meta name="author" content="Erika Felix">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
</head>
```

# Basic Formatting Tags

- <!-- ... --> - Comment
- **<b>** Bold text
- **<i>** Italicized text
- **<u>** Underlined text
- **<br/>** Add a line break to the text
- **<hr/>** Add a line break and header rule
- **<p>** Paragraph
- **<h1>, <h2>, <h3>, <h4>, <h5>, <h6>** headers

```html
<html>
<head>
    <title>Özyeğin University</title>
    <meta charset="UTF-8">
</head>
<!-- This a sample page -->
<body>
    <h1><i>hello everybody</i></h1>
    <p align="center">welcome to our first class
</p>
    <h2> what is <u>html</u>?
    </h2>
    <p>
        let start with learning html tags
    </p>
</body>
</html>
```



# HTML Lists

- HTML has tags that output text in a list format
  - <ul> - Unordered (bullet) list
  - <ol> - Ordered (numbered) list
  - <li> - Start a new entry in a list (ordered or unordered)

```
Shopping list:
<ul>
   <li>Oranges</li>
   <li>Bananas</li>
   <li>Bread</li>
</ul>
```

Shopping list:

- Oranges
- Bananas
- Bread

Shopping list:

1. Oranges
2. Bananas
3. Bread

```
Shopping list:
<ol>
   <li>Oranges</li>
   <li>Bananas</li>
   <li>Bread</li>
</ol>
```

# HTML Links

- Create a hyperlink using the <A> tag
- The most important attribute is the HREF attribute
  - Used to specify the URL of the location to link to
- The link can refer to an HTML page, a servlet, an image, ...

```
<a href="https://www.ozyegin.edu.tr/"> Özyeğin Üniversitesi</a>
```

Özyeğin Üniversitesi

# Table Tags

- A table is specified by providing tags for each row; the columns are specified with each row
- Tags:
  - **<table>** - new table
  - **<tr>** - new row
  - **<td>** - new cell within a row
  - **<th>** - heading cell within a table

```
<table border="1">
    <tr>
        <th>Student</th>
        <th>ID</th>
        <th>Score</th>
    </tr>
    <tr>
         <td>Erika Smith</td>
         <td>12450</td>
         <td>87</td>
    </tr>
    <tr>
         <td>Linda Brown</td>
         <td>34545</td>
         <td>95</td>
    </tr>
</table>
```

| Student | ID | Score |
|---|---|---|
| Erika Smith | 12450 | 87 |
| Linda Brown | 34545 | 95 |

# The Image Tag

- Image tags are used to display graphical images. Some attributes:
- src :
  - source URL of the image
- alt:
  - alternative text to display if the browser can't (or won't) display the graphic
- height , width :
  - Used to customize the size of the image without altering the source file

```
<img src="https://www.ozyegin.edu.tr/sites/all/themes/basic/images/footerlogoTR.png"
    alt="Özyeğin Üniversitesi"
    width="150"
    height="50">
```

# Html Forms

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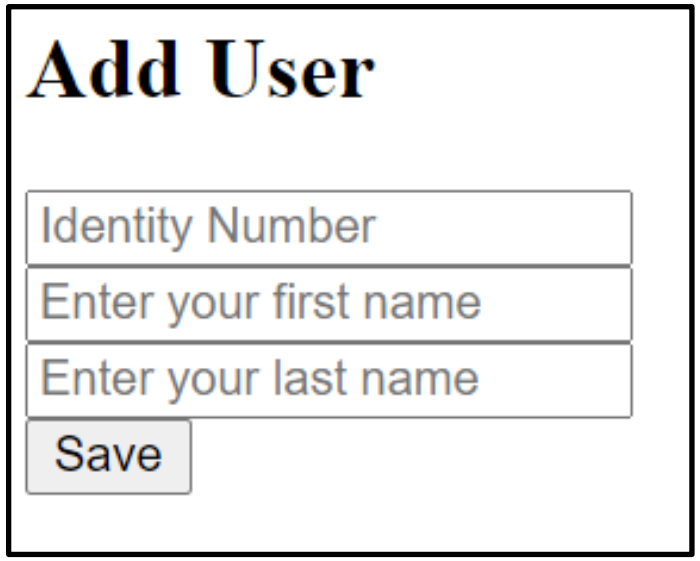
# The Form Tag

- An HTML form is used to collect user input.
- The user input is most often sent to a server for processing.
- Form tags have an **action**
  - Specifies a target URL that is accessed when the form is “submitted”
  - The values for all form elements are packaged and sent to the target
- A web page may contain multiple forms
  - The actions for each form may be the same or different

# Example




```
<h2>Add User</h2>
<form method="POST" action="http://localhost:3000/students">
    <input name="id" type="text" placeholder="Identity Number">
    <input name="firstname" type="text" placeholder="Enter your first name">
    <input name="lastname" type="text" placeholder="Enter your last name">
    <button type="submit">Save</button>
</form>
```

# Form Attributes

- action
  - The target URL that will process the form data
- method
  - How to perform action
  - Determines how the form data is transferred to the target on the server
- target
  - specifies where to display the response that is received after submitting the form
    - _blank : The response is displayed in a new window or tab
    - _self: The response is displayed in the current window (default)
    - _parent : The response is displayed in the parent frame
    - _top The response is displayed in the full body of the window

# Get or Post?

## GET

- This is the default method
- Appends the form data to the URL, in name/value pairs
  - NEVER use GET to send sensitive data! (the submitted form data is visible in the URL!)
- The length of a URL is limited (2048 characters)
- Useful for form submissions where a user wants to bookmark the result
- GET is good for non-secure data, like query strings in Google

## POST

- Appends the form data inside the body of the HTTP request (the submitted form data is not shown in the URL)
- POST has no size limitations and can be used to send large amounts of data.
- Form submissions with POST cannot be bookmarked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# HTML Form Elements and Attributes

ÖZYEĞİN UNIVERSITY

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# Form Elements

- All form elements share certain characteristics
  - The “name” is used to specify the name of the element
  - The “value” specifies the value of the element
    - provides a default value
- The “type” specifies the type of the element
  - e.g., text, checkbox, radio, list, etc.

```
<input type="text" name="firstname" value="Ahmet">
```

# Frequently Used Form Elements

- <input>
- <datalist>
- <label>
- <textarea>
- <button>
- <fieldset>
- <select>
- <option>
- <optgroup>



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# <input> element

- Most used form element
- Can be displayed in several ways, depending on the type attribute

```
<input type="text" name="name" value="John Smith">
<input type="password" name="pwd">
<input type="submit" value="Submit">
```

# Input Types

- <input type="button">
- <input type="checkbox">
- <input type="color">
- <input type="date">
- <input type="datetime-local">
- <input type="email">
- <input type="file">
- <input type="hidden">
- <input type="image">
- <input type="month">
- <input type="number">
- <input type="password">
- <input type="radio">
- <input type="range">
- <input type="reset">
- <input type="search">
- <input type="submit">
- <input type="tel">
- <input type="text">
- <input type="time">
- <input type="url">
- <input type="week">



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# <label> element

- Defines a ***label*** for form elements
- The ***for*** attribute of the <label> tag should be <u>equal to the ***id*** attribute</u> of the <input> element to bind them together.

```
<label for="fname">First name:</label>
<input id="fname" name="firstname" type="text" value="">
```

First name:

# <select> element

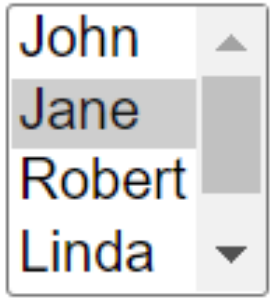
- List permits one or more “options” to be selected
  - The “size” indicates the number of visible lines
  - The “value” for the selected item is provided to your backend

```
<select name="friend" size="4" multiple>
    <option value="1">John</option>
    <option value="2" selected >Jane</option>
    <option value="3">Robert</option>
    <option value="4">Linda</option>
    <option value="5">Erika</option>
</select>
```

select more than one value can be selected

number of visible values

pre-selected



# <textarea> element

- Defines a multi-line input field

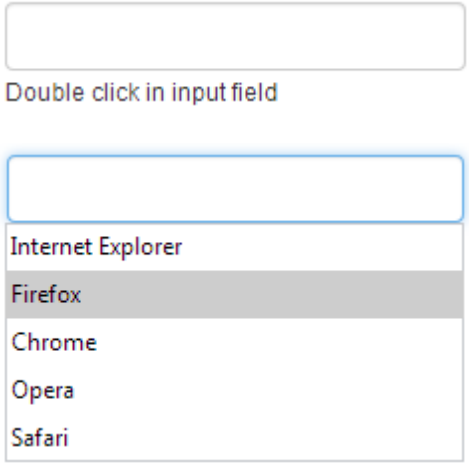
```
<label for="address">Address:</label>
<textarea id= "address" name="address" rows="10" cols="30">
</textarea>
```

Özyeğin University, Çekmeköy Campus
Nişantepe District, Orman Street, 34794 Çekmeköy - İSTANBUL

# <datalist> element

- provide an "autocomplete" feature for an <input> element

```
<form>
   <input list="browsers" name="browser" />
   <datalist id="browsers">
       <option value="Internet Explorer">
       <option value="Firefox">
       <option value="Chrome">
       <option value="Opera">
       <option value="Safari">
   </datalist>
   <input type="submit" />
</form>
```



# <button> element

- defines a clickable button

```
<button type="button" onclick="alert('Hello World!')">Click Me!</button>
```

```
<button type="submit" value="submit">Register Me</button>
```

# Some important input types



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# <input type="text">

- Defines a single-line text input field:
  - The "value" specifies what is returned

```
<input type="text" name= "city" value="İstanbul">
```

İstanbul

# <input type="checkbox">

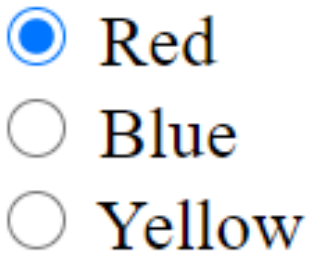
- Checkboxes gather *boolean* information
  - The "value" specifies what is returned if the checkbox is selected
  - The checkbox is initially selected if the checked="true" assigned

```
<label for="3d">I want to pay with 3D Secure</label>
<input id="3d" type="checkbox" name="3dsecure" value="3D Secure" checked="true">
```

I want to pay with 3D Secure ☑

# <input type="radio">

- Defines a radio button
- User can select ONLY ONE of a limited number of choices
- A collection of mutually-exclusive radio buttons share the same “name”
  - The “checked” tag indicates the default value

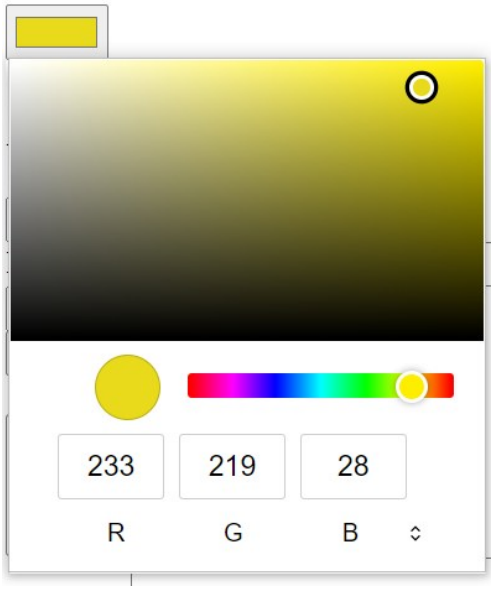



```
<input type="radio" id="red" name="color" value="red" checked="checked">
<label for="red">Red</label><br>
<input type="radio" id="blue" name="color" value="blue">
<label for="female">Blue</label><br>
<input type="radio" id="yellow" name="color" value="yellow">
<label for="yellow">Yellow</label><br>
```

# <input type="color">

- Select a color from a color picker:

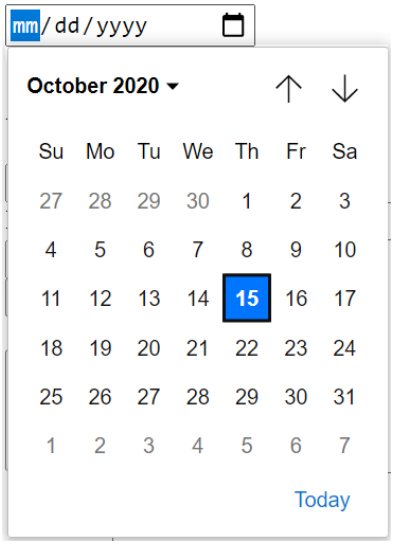
```
<input type="color" name="color" />
```



# <input type="date">
# <input type="time">

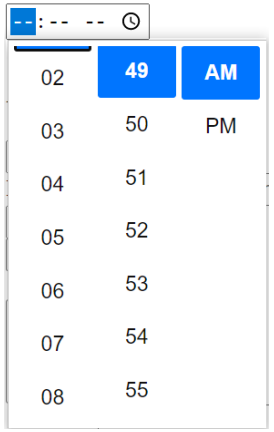
- <date> : allows the user to select a date

```
<input type="date" name="birthday" />
```




- <time> : allows the user to select a time

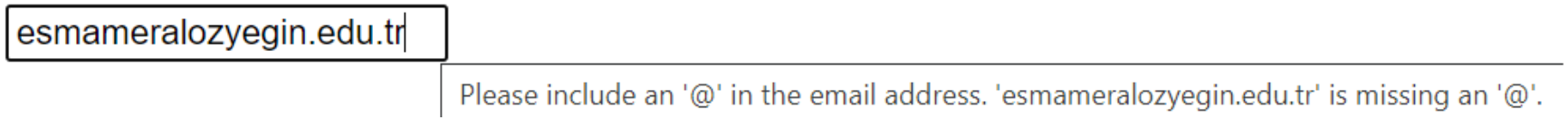
```
<input type="time" name="trxdate" />
```



# <input type="email">

- Used for input fields that should contain an e-mail address.

```
<input type="email" name="email" />
```



# <input type="number">
# <input type="range">

- <number>: input fields that should contain a numeric value

```
<label for="grade">Grade:</label>
<input type="number" id="grade" name="grade" min="1" max="100">
```

Grade: 89

- <range>: defines a control for entering a number whose exact value is not important (like a slider control).

Volume

```
<label for="range">Grade:</label>
<input  id="range" type="range" name="range" min="1" max="10" /> Volume
```

# <input type="submit">

- defines a button for submitting form data

```
<form ...>
    <input name="id" type="text" placeholder="Identity Number">
    <input name="firstname" type="text" placeholder="Enter your first name">
    <input name="lastname" type="text" placeholder="Enter your last name">
    <input type="submit">Submit</button>
</form>
```

Identity Number

Enter your first name

Enter your last name

Submit

# Notes About Submitting Forms…

- The submit event fires when the user clicks a **submit** button or **presses Enter** while editing a field (e.g. <input type="text">) in a form.
  - These 2 items do the same job:

```
<form ...>
   <input type="submit" value="Calculate Age">
   <button type="submit">Find My Age</button>
</form>
```

# …Notes About Submitting Forms

- <input type="submit"> and <button type="submit"> should **be used in forms** only and they will send a request (either GET or POST) to specified URL.
- <input type="button"> can be used anywhere, not just within form and they do not submit form if they are in one.
- The difference is that <button> can have content, whereas <input> cannot
- If you return false from an **onsubmit** event handler, this will prevent the submit of the form to the server from happening

# For More Info and Examples

w3schools.com

https://www.w3schools.com/html
https://www.w3schools.com/html/html_examples.asp

tutorialspoint
SIMPLYEASYLEARNING

https://www.tutorialspoint.com/html/index.htm
https://www.tutorialspoint.com/html5/index.htm

# Any Question?

***Thank you for your attention***

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